Audit of the Full-Covariance Gamma-Method Implementation

What is established, what the code computes, and what remains unvalidated

Technical note for GradientFlowRGToolkit

31 August 2026

Current scientific status: experimental. This note audits the estimator now named `ExperimentalRectangularLongRunCovariance`; it is no longer the primary estimator. The adopted Bartlett/Newey–West implementation and its validators are documented in `long-run-covariance-methods.pdf`. The full long-run covariance matrix is a standard quantity already defined in Wolff's Gamma-method paper. That target was not invented by the agent that wrote the legacy code. However, the legacy implementation's window-selection rule, mandatory minimum window, separate-channel estimation, and eigenvalue clipping do not together constitute a literature-validated implementation of Wolff's method. A scalar independent-data test demonstrates substantial finite-sample bias. The current estimator should therefore not yet be used as authoritative statistical evidence.

1 The essential distinction

Two claims must not be conflated:

1. **The target is justified.** For a stationary vector history, the covariance of the sample mean depends on the full matrix of lagged cross-correlations. This is established statistical theory.
2. **The particular estimator is not yet justified.** There are many ways to truncate, taper, regularize, or otherwise estimate that infinite lag sum. The current code makes several choices that do not match Wolff's published practical prescription and have not otherwise been validated.

The legacy implementation in `ContinuousBetaFunction` was generated by a coding agent. Its presence in that codebase is provenance, not scientific validation. The present audit therefore starts again from the primary literature and from falsifiable synthetic tests.

2 The full-matrix target is standard

Let $X_n \in \mathbb{R}^M$ be a stationary vector-valued Monte Carlo history, with mean μ, and define

$$\Gamma_{ab}(k) = \mathbb{E}\left[(X_{n,a} - \mu_a)(X_{n+k,b} - \mu_b)\right].$$

The long-run covariance, also called the zero-frequency spectral covariance, is

$$\Sigma = \sum_{k=-\infty}^{\infty} \Gamma(k) = \Gamma(0) + \sum_{k=1}^{\infty} \left[\Gamma(k) + \Gamma(k)^{\mathsf{T}}\right].$$

Under the usual Markov-chain central-limit conditions,

$$\sqrt{N}(\overline{X}-\mu) \Longrightarrow \mathcal{N}(0,\Sigma), \qquad \mathrm{Cov}(\overline{X}) \simeq \frac{\Sigma}{N}.$$

Wolff defines the matrix-valued $\Gamma_{\alpha\beta}(n)$, the full sum $C_{\alpha\beta} = \sum_t \Gamma_{\alpha\beta}(t)$, and the resulting covariance of the means explicitly in Eqs. (2)–(4) and (12)–(13) of Ref. [1]. Modern multivariate spectral-variance theory uses the same target [2]. Extending beyond diagonal error bars is therefore not a speculative extension of the Gamma method; it restores information already present in its general formulation.

3 What the current code computes

For one channel, the input is an $N \times M$ matrix. Configurations index the rows and retained flow times index the columns. The code first centers each column:

$$\delta X_{na} = X_{na} - \overline{X}_a.$$

It estimates each diagonal normalized autocorrelation using

$$\widehat{\rho}_a(k) = \frac{\sum_{n=1}^{N-k} \delta X_{na}\delta X_{n+k,a}}{\sum_{n=1}^{N} \delta X_{na}^2}.$$

Beginning with $\widehat{\tau}_a = 1/2$, it adds selected positive autocorrelations while $k < S\widehat{\tau}_a$, then chooses one matrix-wide window

$$W = \left\lceil S \max_a \widehat{\tau}_a \right\rceil,$$

forces $W \geq 4$ when the sample length permits, and caps it at $N/4$.

For

$$G_{ab}(k) = \sum_{n=1}^{N-k} \delta X_{na}\delta X_{n+k,b},$$

the code returns the pre-projection estimate

$$\boxed{\widehat{\mathrm{Cov}}(\overline{X}_a, \overline{X}_b) = \frac{1}{N^2}\left[G_{ab}(0) + \sum_{k=1}^{W}\{G_{ab}(k) + G_{ba}(k)\}\right].}$$

This is a multivariate rectangular-window spectral-covariance estimator. It does estimate off-diagonal lag corrections from data rather than inferring them from diagonal autocorrelation times.

4 Why this is not Wolff's published practical algorithm

Wolff's sample autocovariance uses the lag-dependent denominator $N-k$ for one replica, not the constant-N convention implicit above. More importantly, Wolff's automatic procedure defines an effective exponential time and a function $g(W)$, then selects the first sign change of $g(W)$; see Eqs. (50)–(52) of Ref. [1]. The code's $W = \lceil S \max_a \widehat{\tau}_a \rceil$ rule is not that procedure.

Wolff also derives a leading correction for the bias introduced by estimating the mean from the same finite sample; see Eqs. (32) and (49). The current code does not apply it. A maximum over coordinate-wise autocorrelation times is not guaranteed to control the autocorrelation time of every linear combination. Consequently, describing this implementation simply as "following Wolff's method" overstates what the source supports.

5 A decisive independent-data check

Suppose X_n are independent scalar observations with variance σ^2. There is no physical autocorrelation to sum. Nevertheless, the current code normally forces $W = 4$. Because every observation is centered using the sample mean,

$$\mathbb{E}[G(0)] = (N-1)\sigma^2, \qquad \mathbb{E}[G(k)] = -\frac{N-k}{N}\sigma^2 \quad (k > 0).$$

For a fixed window W, the expected pre-projection estimate relative to the true variance of the mean, σ^2/N, is therefore

$$\frac{\mathbb{E}[\widehat{\mathrm{Var}}(\overline{X})]}{\sigma^2/N} = 1 - \frac{2W+1}{N} + \frac{W(W+1)}{N^2}.$$

With $W = 4$, the expected ratio is only 0.60 at $N = 20$. This failure is structural, not a floating-point artifact.

A 20,000-replication standard-normal simulation of the actual implementation gave:

N	mean estimate	relative bias	negative before clipping
20	0.621	−37.9%	14.68%
50	0.823	−17.7%	2.40%
100	0.915	−8.5%	0.16%
400	0.975	−2.5%	approximately zero

The target in every row is one after scaling by N. Eigenvalue clipping raises some negative scalar estimates to zero, explaining the small difference from the analytic pre-projection expectation, but does not repair the bias.

6 Positive-semidefinite projection

A rectangular finite lag sum need not be positive semidefinite. The current implementation diagonalizes the estimate and replaces negative eigenvalues by zero:

$$C = Q\Lambda Q^{\mathsf{T}}, \qquad C_{\mathrm{PSD}} = Q\max(\Lambda, 0)Q^{\mathsf{T}}.$$

For a symmetric matrix, this is the nearest PSD matrix in Frobenius norm [4]. That is an algebraic result, not a statistical validation: it establishes neither bias nor confidence-interval coverage, and it can erase scientifically important low-variance directions.

The adjustment magnitude and affected directions should be recorded whenever projection is used. A small Frobenius change is useful diagnostic evidence, but it does not establish that the projected estimator has valid statistical coverage.

Positive-semidefinite lag-window estimators exist. Bartlett/Newey–West weights, for example, are PSD by construction [3]. This does not remove the need to justify bandwidth selection or validate finite sample performance.

7 Two additional covariance problems

7.1 Correlated channels are currently separated

The raw combined s history correctly includes its covariance with the raw p measurements used to form it. But the code invokes the estimator separately for final p, combined s, and c blocks.

The resulting software objects therefore encode zero covariance between those blocks. When the channels are measured on matched configurations, the statistically correct unit is their jointly concatenated per-configuration vector. This is especially important for p and combined s, because the latter is constructed partly from the former.

7.2 High dimension implies singularity

Any estimator expressible as $X^{\mathsf{T}}KX$ from a centered $N \times M$ history has rank at most $N - 1$. Whenever $M > N$, a valid PSD covariance is therefore necessarily singular. Neither a PSD window nor eigenvalue clipping can create information in unsupported directions. Downstream inverse-covariance fits require an explicit dimension-reduction, low-rank, flow-time-selection, or independently justified regularization policy.

8 What remains trustworthy in the processing chain

Conditional on a valid joint covariance estimate, deterministic propagation through

$$g_{\mathrm{GF}}^2(t) = \mathcal{N}(t)E(t), \qquad \beta_{\mathrm{GF}}(t) = -t\frac{dg_{\mathrm{GF}}^2}{dt}$$

is conceptually appropriate: coupling and beta-function values should retain their inherited correlations. The present concern is upstream—whether the covariance supplied to that propagation is a defensible estimate.

9 Required validation before adoption

The estimator decision should be made explicitly rather than inherited from the legacy agent-generated implementation. At minimum, candidates should pass the following tests:

1. scalar and vector IID data recover Σ/N without a forced-lag bias;
2. a vector AR or VAR process recovers its analytically known long-run covariance;
3. linear equivariance holds: $\widehat{\Sigma}(AX) = A\widehat{\Sigma}(X)A^{\mathsf{T}}$;
4. selected derived scalar histories agree with a faithful projected Wolff calculation;
5. cross-channel covariance is retained from matched configurations;
6. coverage and bandwidth sensitivity are measured, not inferred from PSD alone; and
7. singular high-dimensional output is represented honestly and never silently inverted.

A defensible comparison would include (i) a faithful Wolff calculation for selected scalar derived quantities and (ii) a sourced multivariate spectral estimator, such as Bartlett/Newey–West with an explicit bandwidth policy. Until that comparison is complete, the current rectangular estimator plus projection should be named as an experimental comparator, not frozen in as "the Gamma method."

10 Conclusion

The user's worry identifies a real scientific governance problem: agent-written analysis code must not acquire authority merely by surviving in a legacy repository. In this case, primary sources validate the desired full-matrix quantity, which is reassuring. They do not validate the current estimator's specific combination of heuristics. The independent-data counterexample is sufficient to reject the present implementation as a production default.

References

[1] U. Wolff, "Monte Carlo errors with less errors," *Computer Physics Communications* 156 (2004) 143–153. arXiv:hep-lat/0306017.

[2] D. Vats, J. M. Flegal, and G. L. Jones, "Strong consistency of multivariate spectral variance estimators in Markov chain Monte Carlo," *Bernoulli* 24 (2018) 1860–1909. arXiv:1507.08266.

[3] W. K. Newey and K. D. West, "A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix," *Econometrica* 55 (1987) 703–708. doi:10.2307/1913610.

[4] N. J. Higham, "Computing a nearest symmetric positive semidefinite matrix," *Linear Algebra and its Applications* 103 (1988) 103–118. doi:10.1016/0024-3795(88)90223-6.